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Advanced Training System Designed to Enhance Athletic Performance in Swinging Sports

Abstract

The present invention relates to a training system designed to enhance athletic performance in swinging sports through an innovative approach. Specifically, the system comprises an adjustable tension mechanism configured to connect between a user's shoulder/chest area and a lower extremity, facilitating variable resistance to improve swing mechanics. Additionally, the invention includes an adjustable tee stand capable of height, velocity, and angle adjustments, paired with a dynamic target release mechanism. This mechanism enables controlled movement of a target towards the user, simulating real target flight paths and offering a comprehensive training experience. The system's adaptability offers significant improvements over traditional training aids by providing multi-dimensional feedback, accommodating athletes of various skill levels, and simulating a wide range of athletic scenarios. Its modular design allows for customization to individual needs, promoting effective and efficient training outcomes across multiple swinging sports such as baseball, golf, and tennis.

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Background/Summary

[0001] The present invention relates to the technical field of sports training systems, particularly to an integrated system designed to enhance athletic performance in swinging sports such as baseball, golf, tennis, and the like.

[0002] The realm of athletic training, especially in swinging sports, has seen various technological innovations aimed at improving athletes' performance. Prior art in this domain includes US Patent Application #20210008426, which discloses a swing training system utilizing a shoulder harness with coupling members for developing efficient swing mechanics. Similarly, US Patent Application #20230191225 presents a rotation swing training system focused on body rotation analysis using electronic feedback. Additionally, U.S. Pat. No. 8,668,602 introduces a system for baseball and softball training using a target object projection to provide multi-dimensional swing feedback.

[0003] These existing solutions, while innovative, exhibit limitations in providing a fully comprehensive training system that addresses the multifaceted nature of athletic swinging performance. For instance, they may focus narrowly on specific aspects of the swing, such as body rotation or swing plane, without integrating feedback across multiple dimensions of swing mechanics. Moreover, the reliance on static training aids or specific feedback mechanisms limits their applicability across various sports and athlete skill levels.

[0004] There exists a significant gap in the current technology landscape for a versatile and integrated training system capable of addressing the comprehensive needs of athletes in swinging sports. This gap underscores the need for a system that not only provides multi-dimensional feedback on swing mechanics but also offers adjustable resistance, dynamic target engagement, and customizable training routines to accommodate athletes at different stages of skill development.

[0005] The objectives of the present invention are to: provide an integrated training system that enhances athletic performance in swinging sports through adjustable tension mechanism and dynamic target engagement; offer a versatile solution applicable to various swing based sports and athlete skill levels by incorporating customizable components and feedback mechanisms to address limitations of existing training aids; provide a comprehensive suite of tools that support both linear and rotational swing phases, facilitating a complete training experience.

[0006] The proposed invention offers significant improvements over existing solutions by integrating various components into a single system. The proposed invention offers significant improvements over existing solutions via an adjustable tension mechanism facilitating swing guidance, and an adjustable target mechanism to simulate various projectile trajectories for the user.

[0007] This system allows for a holistic training approach that can be tailored to an individual athlete's needs, supporting a broader range of swing training scenarios and offering a more effective means of enhancing athletic performance in swinging sports. Through its innovative design, the invention presents a versatile and comprehensive training tool that bridges the gap identified in the current technology landscape.

[0008] By addressing these unmet needs and surpassing the limitations of related art, the invention sets a new standard in athletic performance training, offering a path to achieving optimal swing mechanics and enhancing overall athletic capabilities in swinging sports.

SUMMARY

[0009] In view of the circumstances outlined above, aspects of the present invention discloses systems and methods for the implementation of an advanced training system designed to enhance athletic performance in swinging sports through a comprehensive approach that integrates an

adjustable tee stand composed of a rope, sliding rails, a structural frame, a base, a pole and a ball; and an adjustable tension mechanism comprising: wearable attachment points, one or more lower extremity harness, a chest-shoulder harness, a waist harness and a stretch band. Situated within the technical field of sports training systems, it aims to provide athletes with a versatile tool for improving swing mechanics across a variety of sports, including baseball, golf, tennis, and others. [0010] Traditional training aids for swinging sports often focus on isolated aspects of swing mechanics and lack the ability to offer holistic, multi-dimensional feedback. This invention addresses this limitation by offering an integrated solution that enables adjustable/dynamic resistance, dynamic target interaction, and customizable training mechanisms, thereby presenting a novel approach to athletic training in swinging sports.

[0011] According to one aspect of the invention, there is provided a training system for enhancing athletic performance in swinging sports, comprising: an adjustable tension mechanism that connects between a user's shoulder area and a lower extremity, offering variable resistance to enhance both linear and rotational swing phases; a lower extremity harness for anchoring the tension mechanism to the user's foot or lower extremity area, ensuring stability and support during dynamic movements, as well as a corresponding chest harness facilitating a secondary anchor point for the tension mechanism; an adjustable tee stand capable of height and angle adjustments, facilitating a comprehensive range of swing training scenarios; and a dynamic target release mechanism associated with the tee stand, enabling controlled movement of a target towards the user to simulate real target flight paths.

[0012] This training system offers significant advantages over prior art, including enhanced versatility to accommodate athletes at different skill levels, comprehensive feedback on swing mechanics, and the ability to simulate various athletic scenarios. The system's modular design allows for customization to individual needs, promoting more effective and efficient training outcomes.

[0013] The invention's versatility extends its applicability across multiple swinging sports, making it an ideal training aid for athletes seeking to improve their performance in baseball, golf, tennis, and similar sports. Its comprehensive approach to swing training also makes it suitable for use in educational and professional sports training facilities, offering a tool that can adapt to various training requirements and athlete development programs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1A depicts the adjustable tee stand for the athletic training performance system.

[0015] FIG. 1B depicts the adjustable tee stand for the athletic training performance system after the system has been activated.

[0016] FIG. 1C depicts the adjustable tee stand for the athletic training performance system from a lateral view.

[0017] FIG. 2A depicts components and configuration of the adjustable tension mechanism.

[0018] FIG. 2B-E depicts various embodiments of the adjustable tension mechanism. Cl

DETAILED DESCRIPTION

[0019] FIG. 1A presents the initial setup of the adjustable tee stand, demonstrating the integration and relative positioning of its primary components. The adjustable tee stand (10) is viewed from the front, where the base (14) is positioned at a starting point along the sliding rail (12). This configuration establishes the fundamental setup for the system, ready for user interaction.

[0020] The pole (16) is depicted extending vertically from the base (14), holding the ball (18) aloft in a ready position. The structural frame (20) guides the rope (22), which can be seen depicted in its primed state resisting the gravitational force acting on the base mechanism.

[0021] The sliding rail (12) is constructed to guide the base (14) to move along its length toward the user, offering the adjustability needed to cater to different ball movement trajectories. The tension of the rope (22) can be manipulated by the user, wherein the user can control the release mechanism for the ball (18). This setup is pivotal for simulating various pitch speeds and trajectories as well as providing a realistic training experience.

[0022] FIG. 1B, the athletic performance training system is shown after the rope (22) has been released. The base (14) has been moved forward along the sliding rail (12) due to gravitational forces, in relation to the structural frame (20), illustrating the end point of the system post user activation. This movement of the base (14) suggests that the user has since initiated the mechanism. Subsequently the base moves down the sliding rail (12) towards the user simulating the trajectory of an incoming ball from an opposing sports player.

[0023] FIG. 1C depicts a lateral view of the adjustable tee stand illustrated in FIG. 1A and FIG. 1B. In this view the stop pad (24) is depicted. The stop pad (24) catches the base (14) after the system has been activated and the base (14) has slid down the rail (12).

[0024] In various embodiments the adjustable tee stand can be operated with steps comprising: placement of the ball (18) on the pole (16); adjustment of the sliding rails (12); priming of the system via implementing tension on the rope (22) attached to the base (14); movement of the base (14) to the primed position; the release of the rope (22); the movement of the base (14) towards the user due to gravity via the sliding rails (12); and the resultant simulation of a flight path of the ball (18) towards the user creating an opportunity for the user to practise a swing.

[0025] FIG. 2A depicts the adjustable tension mechanism (30) during the pre-swing phase and its corresponding components used to train the athlete's swing motion. The tension mechanism has components comprising: a lower extremity harness (34) built with an adjustable strap to fit securely around the user's anterior lower extremity; a shoulder/chest harness (32) attached to the user's upper body; and a stretch band (36) connecting the lower extremity harness to the shoulder/chest harness via wearable attachment points. These components represent the core of the adjustable tension mechanism.

[0026] Embodiments of the tension mechanism substitute various materials for the stretch band (36) comprising ropes, rubber bands, or elastics, each selected for its specific elasticity and tension properties. The stretch band is central to the system's ability to provide feedback and resistance, guiding the user to activate an optimal combination and sequence of muscles for swing performance.

[0027] FIG. 2B depicts the adjustable tension mechanism (30) during the post-swing phase. In this embodiment the stretch band (36) has been activated creating tension during the swing motion consequently providing feedback to the user to help guide the swing. Surplus stretch band (38) is depicted freely hanging from the back of the shoulder/chest harness (32). This surplus stretch band (38) is an extension of the stretch band (36) allowing a separate person or coach to dynamically adjust the tension in the stretch band (36) by pulling on the surplus stretch band (38). The dynamic tension adjustment from the separate person or coach further improves swing training by enabling the person or coach to provide dynamic feedback assisting in swing guidance.

[0028] FIG. 2C depicts an embodiment of the adjustable tension mechanism integrating the same features as previous embodiments. The variation in this embodiment is featured in the waist/torso harness (32) which has been repositioned from the shoulder/chest configuration in FIG. 2B. The opposing end of the stretch band (36) is anchored on the anterior or leading lower extremity of the user via a lower extremity harness (34). This embodiment changes the tension angles provided by the system helping to guide and improve torso/waist movement during swing motion. Additionally, this embodiment continues to feature surplus stretch band (38) for dynamic tension adjustments from a separate person or coach.

[0029] FIG. 2D depicts an embodiment of the adjustable tension mechanism integrating the same features as previous embodiments. The variation in this embodiment is featured in the second lower

extremity harness (32) which has been repositioned from the shoulder/chest configuration in FIG. 2B. The opposing end of the stretch band (36) is anchored on the anterior lower extremity of the user via a lower extremity harness (34). This embodiment modifies the tension angles provided by the system helping to guide, position and improve lower extremity movement during swing motion. [0030] FIG. 2E depicts an embodiment of the adjustable tension mechanism illustrated in FIG. 2C wherein the lower extremity harness (34) in this embodiment has been positioned on the anterior or trailing lower extremity to guide and position the torso/waist of the user with respect to the anterior or trailing lower extremity during swing motion.

[0031] Each part of the tension system can be used individually or it can be used collectively as a group. Coaches can also effect the pulling of the rope.

[0032] The embodiments described above are given for the purpose of facilitating the understanding of the present invention and are not intended to limit the interpretation of the present invention. The respective elements and their arrangements, materials, conditions, shapes, sizes, or the like of the embodiment are not limited to the illustrated examples but may be appropriately changed. Further, the constituents described in the embodiment may be partially replaced or combined together.

Claims

1. A training system for enhancing athletic performance in swinging sports, comprising: an adjustable tension mechanism designed to connect a user's shoulder/chest area with and a lower extremity; a wearable shoulder/chest harness designed for securing said tension mechanism to the user; a lower extremity harness intended to anchor the opposing end of the adjustable tension mechanism to the user's lower extremity; an adjustable tee stand capable of height, velocity and angle adjustments; and a dynamic target release mechanism associated with the tee stand, enabling movement of a target towards the user in a controlled manner.
2. The system of claim 1, wherein the adjustable tension mechanism can be constructed from alternatives comprising ropes, rubber bands, elastics, and stretch bands.
3. The system of claim 1, wherein the system is configured to allow another person or coach to dynamically vary the tension in the stretch band.
4. The system of claim 1, wherein the harnesses feature wearable attachment points adaptable to a variety of mechanisms via clips, pins, magnets, and hook and loop fasteners, facilitating quick adjustments and secure attachment during use.
5. The system of claim 1, wherein any of the aforementioned harnesses comprise adjustable strapping to accommodate different user sizes.
6. The system of claim 1, wherein the adjustable tension mechanism can be adapted to various harness configurations comprising: shoulder/chest-anterior lower extremity, shoulder/chest-posterior lower extremity, waist/torso-posterior lower extremity, waist/torso-anterior lower extremity, posterior lower extremity-anterior lower extremity.
7. The system of claim 1, wherein the adjustable tee stand includes a mechanism for height adjustment that allows for the tee to be positioned at various heights.
8. The system of claim 1, wherein the dynamic target release mechanism is operable via a selection from the group consisting of a foot-activated lever, a manually operated string rope, and an electronically controlled release system.
9. The system of claim 1, wherein the adjustable tee stand is integrated with a track system that guides the movement of the target along various predetermined paths.
10. The system of claim 1, wherein the target release mechanism includes a variable speed control feature, allowing the speed at which the target moves towards the user to be adjusted.
11. The system of claim 1, including an electronic control unit configured to programmatically adjust the height, velocity, and angle of the tee stand.

- 12.** The training system of claim 1, wherein the pole or tee can be optionally attached or detached.
 - 13.** The system of claim 1, further wherein multiple adjustable tension mechanisms can be used simultaneously to further guide and improve the user's swing motion.
 - 14.** A method for improving swing mechanics in athletes, involving: providing a tension-based feedback loop through a connection between the user's upper body and lower extremity; employing a dynamic target to simulate varying target movement scenarios; adjusting the tension of the feedback loop and-or the aspects of the dynamic target's flight path based on the user's development needs to simulate and improve swing mechanics.
 - 15.** The method of claim 14, wherein adjusting the tension in the tension-based feedback loop involves changing the length or material used to implement it.
 - 16.** The method of claim 14 wherein, the tension-based feedback loop is configured to allow another person or coach to dynamically vary the tension experienced by the user.
 - 17.** The method of claim 14, wherein the tension-based feedback loop can be adapted to various configurations comprising: chest-anterior lower extremity, chest-posterior lower extremity, waist/torso-posterior lower extremity, waist/torso-anterior lower extremity, posterior lower extremity-anterior lower extremity.
 - 18.** The method of claim 14, wherein multiple tension-based feedback loops can be utilized simultaneously.
 - 19.** The method of claim 14, wherein employing a dynamic target involves adjusting the height and angle of a tee stand to simulate different flight locations and trajectories.
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